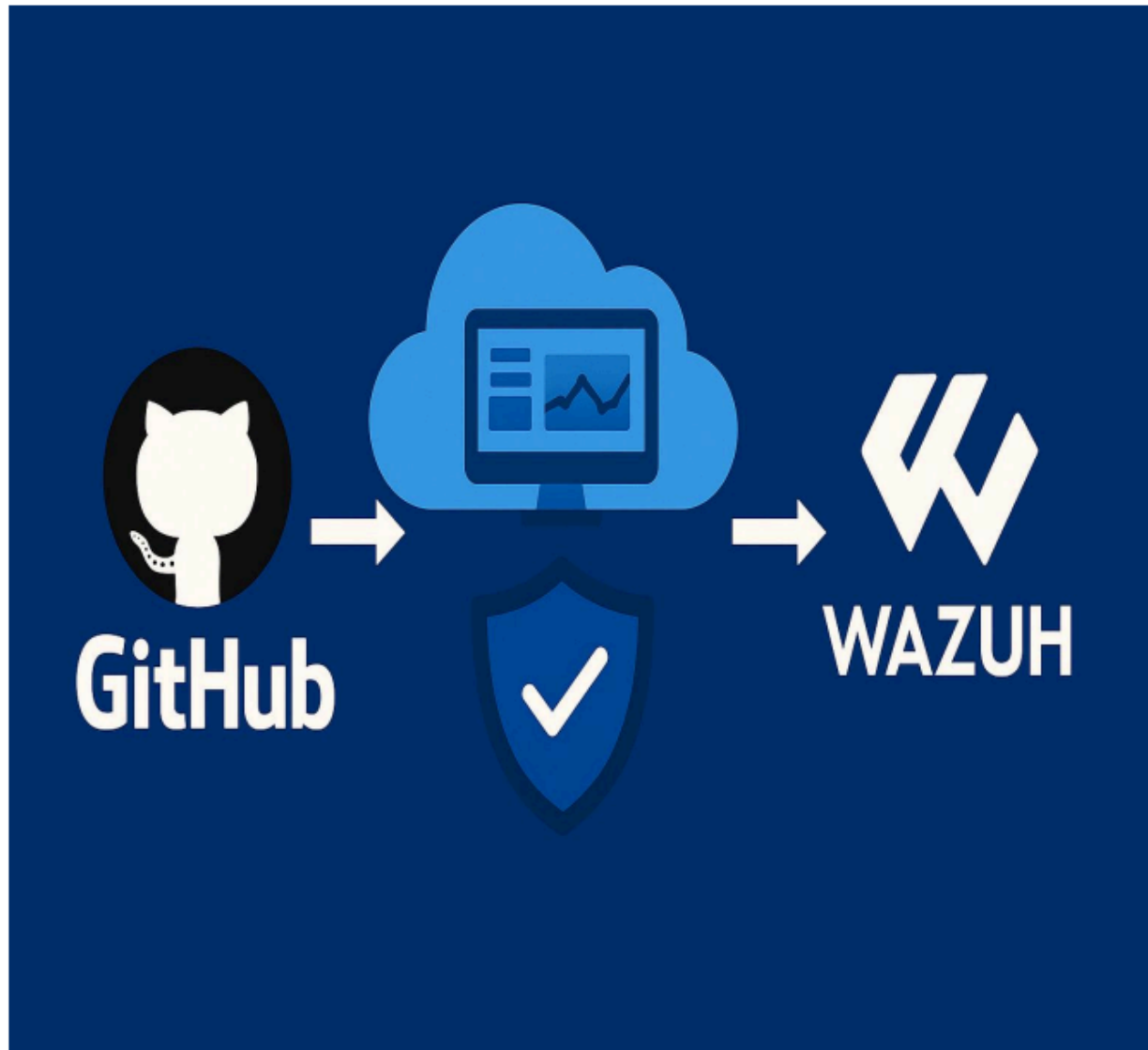


Wazuh Integration with GitHub



Integrating Wazuh with GitHub allows for enhanced security monitoring and compliance checks of your repositories. This integration can help detect and respond to potential security threats and vulnerabilities in your codebase and ensure that your repositories adhere to security policies and best practices.

Key Features:

Monitoring and Alerting: Wazuh can monitor GitHub repositories for changes, including new commits, pull requests, and issues. It can alert security teams about suspicious activities, such as unauthorized access or potentially malicious code changes.

Incident Response: In the event of a security incident, Wazuh can automatically trigger incident response actions. For example, it can revoke access tokens, restrict repository access, or notify the security team for further investigation.

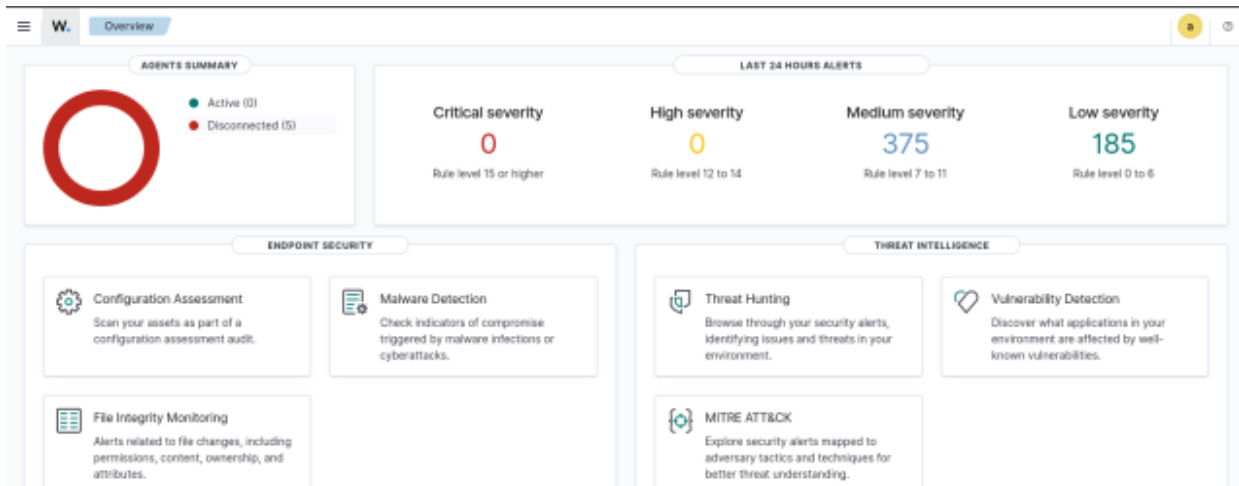
Audit and Reporting: Wazuh provides detailed logs and reports on activities within GitHub repositories, making it easier to audit changes and demonstrate compliance with regulatory requirements.

Here is my Wazuh server running on virtualbox.

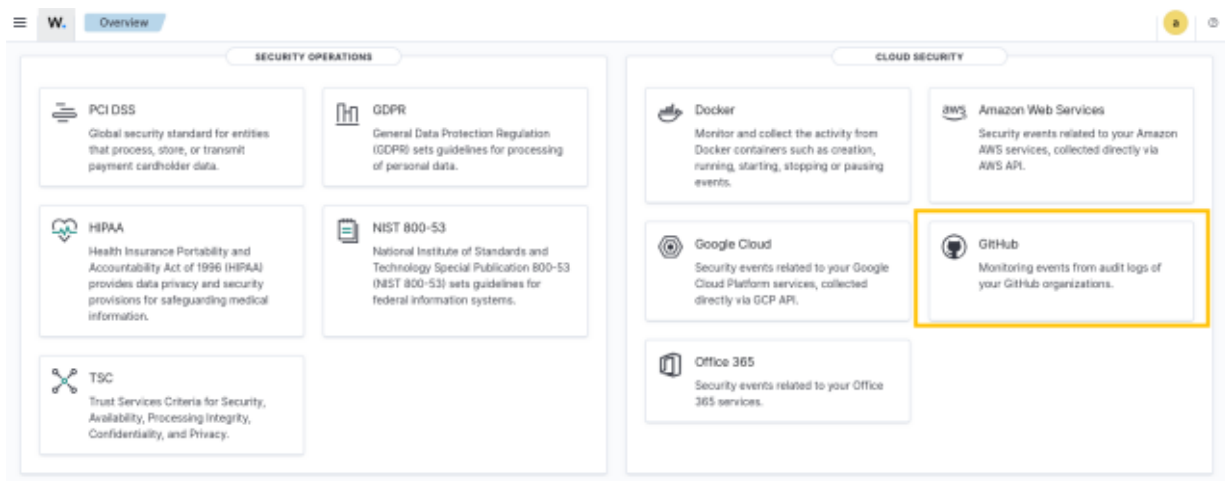
```
(kali@kali) [~/Desktop]
$ sudo systemctl status wazuh-manager
● wazuh-manager.service - Wazuh manager
   Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; preset: disabled)
   Active: active (running) since Sun 2025-07-20 02:46:51 EDT; 11min ago
     Invocation: 91d863e72b784759bfad1c08df071a04
       Process: 110875 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
        Tasks: 217 (limit: 9207)
      Memory: 1.8G (peak: 1.8G)
      CPU: 4min 6.314s
     CGroup: /system.slice/wazuh-manager.service
             └─110965 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
               └─110966 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                 └─110967 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                   └─110970 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                     └─110973 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                       └─110996 /var/ossec/bin/wazuh-integratord
                         └─111017 /var/ossec/bin/wazuh-authd
                           └─111042 /var/ossec/bin/wazuh-db
                             └─111075 /var/ossec/bin/wazuh-execd
                               └─111110 /var/ossec/bin/wazuh-analysisd
                                 └─111131 /var/ossec/bin/wazuh-syscheckd
                                   └─111144 /var/ossec/bin/wazuh-remoted
                                     └─111146 /var/ossec/bin/wazuh-remoted
                                       └─111192 /var/ossec/bin/wazuh-logcollector
                                         └─111207 /var/ossec/bin/wazuh-monitord
                                           └─111229 /var/ossec/bin/wazuh-modulesd

Jul 20 02:46:46 kali env[110875]: Started wazuh-syscheckd ...
Jul 20 02:46:47 kali env[110875]: Started wazuh-remoted ...
Jul 20 02:46:47 kali env[111190]: 2025/07/20 02:46:47 wazuh-logcollector: INFO: (1905): No file configured to monitor.
Jul 20 02:46:47 kali env[110875]: Started wazuh-logcollector ...
Jul 20 02:46:48 kali env[110875]: Started wazuh-monitord ...
Jul 20 02:46:48 kali env[111227]: 2025/07/20 02:46:48 wazuh-modulesd:router: INFO: Loaded router module.
Jul 20 02:46:48 kali env[111227]: 2025/07/20 02:46:48 wazuh-modulesd:content_manager: INFO: Loaded content_manager module.
Jul 20 02:46:49 kali env[110875]: Started wazuh-modulesd ...
Jul 20 02:46:51 kali env[110875]: Completed.
Jul 20 02:46:51 kali systemd[1]: Started wazuh-manager.service - Wazuh manager.
```

Wazuh Dashboard



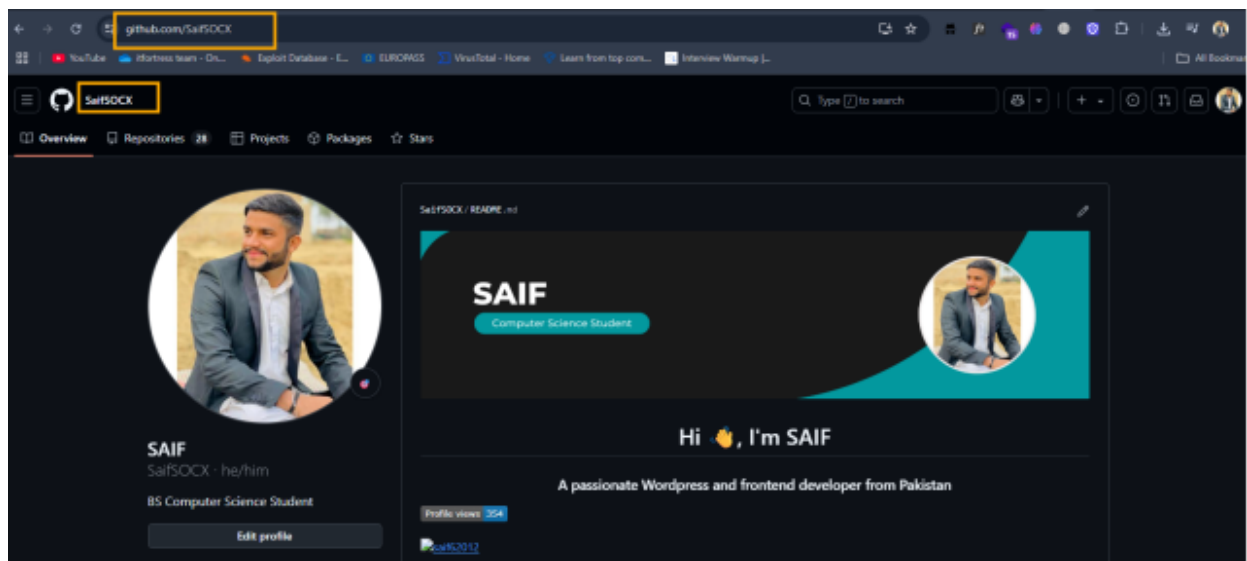
Scroll down in Wazuh-dashboard and go to “Cloud Security” and select “Github” for integration. Make sure you have an account on GitHub.



After selecting there is no logs by default.

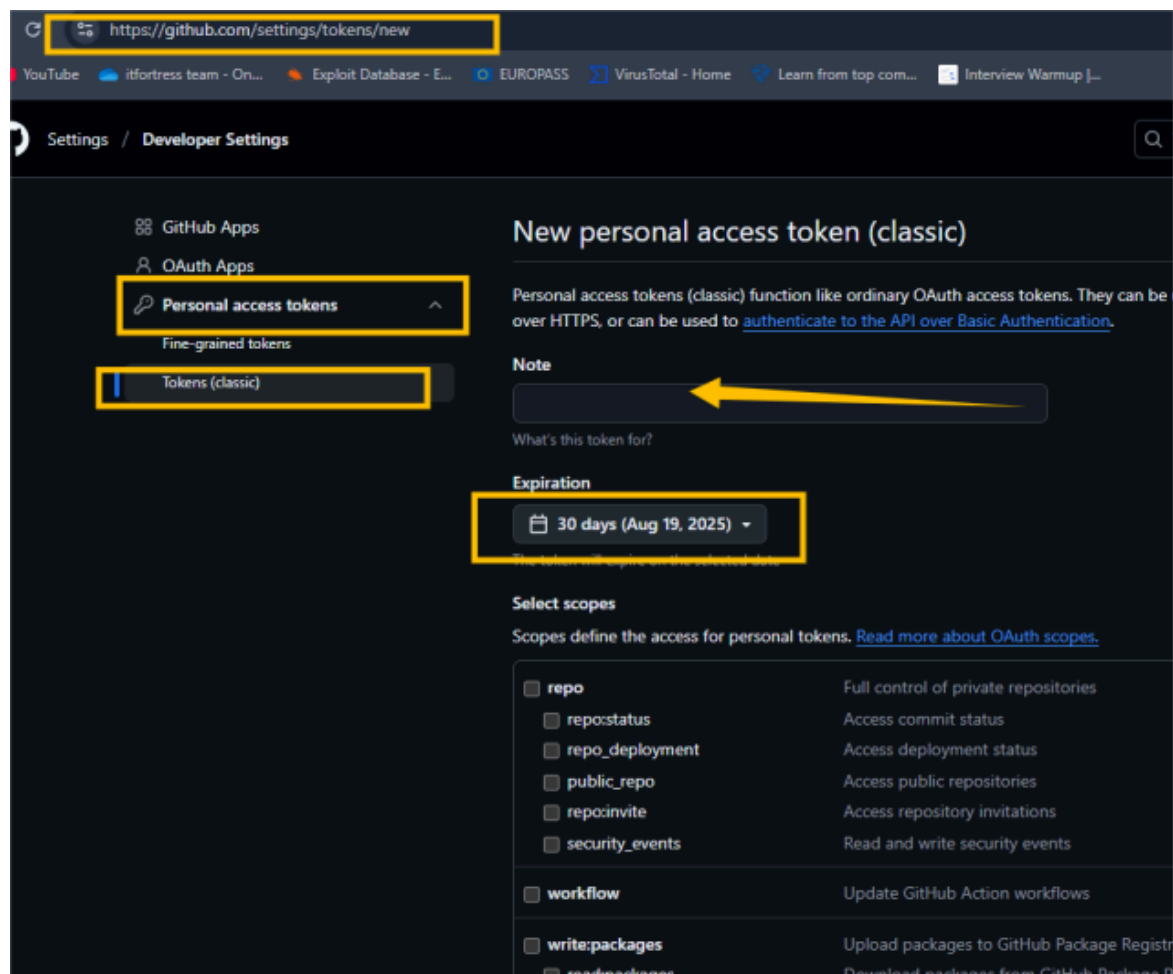


Now login to GitHub account. I am using my own account for integration.



You can visit this link: <https://github.com/settings/tokens/new> for generate new token for Wazuh integration.

<https://github.com/settings/tokens/new>



You can select “Expiration” and “Scop”.

Select scopes

Scopes define the access for personal tokens. [Read more about OAuth scopes.](#)

☐ repo	Full control of private repositories
☐ repo:status	Access commit status
☐ repo_deployment	Access deployment status
☐ public_repo	Access public repositories
☐ repo:invite	Access repository invitations
☐ security_events	Read and write security events
☐ workflow	Update GitHub Action workflows
☐ write:packages	Upload packages to GitHub Package Registry
☐ read:packages	Download packages from GitHub Package Registry
☐ delete:packages	Delete packages from GitHub Package Registry
☐ admin:org	Full control of orgs and teams, read and write org projects
☐ write:org	Read and write org and team membership, read and write org projects
☐ read:org	Read org and team membership, read org projects
☐ manage_runners:org	Manage org runners and runner groups
☐ admin:public_key	Full control of user public keys
☐ write:public_key	Write user public keys
☐ read:public_key	Read user public keys
☐ admin:repo_hook	Full control of repository hooks
☐ write:repo_hook	Write repository hooks
☐ read:repo_hook	Read repository hooks
☐ admin:org_hook	Full control of organization hooks

Creating new “token” for Wazuh integration.

Note

Wazuh

What's this token for?

Expiration

30 days (Aug 19, 2025) ▾

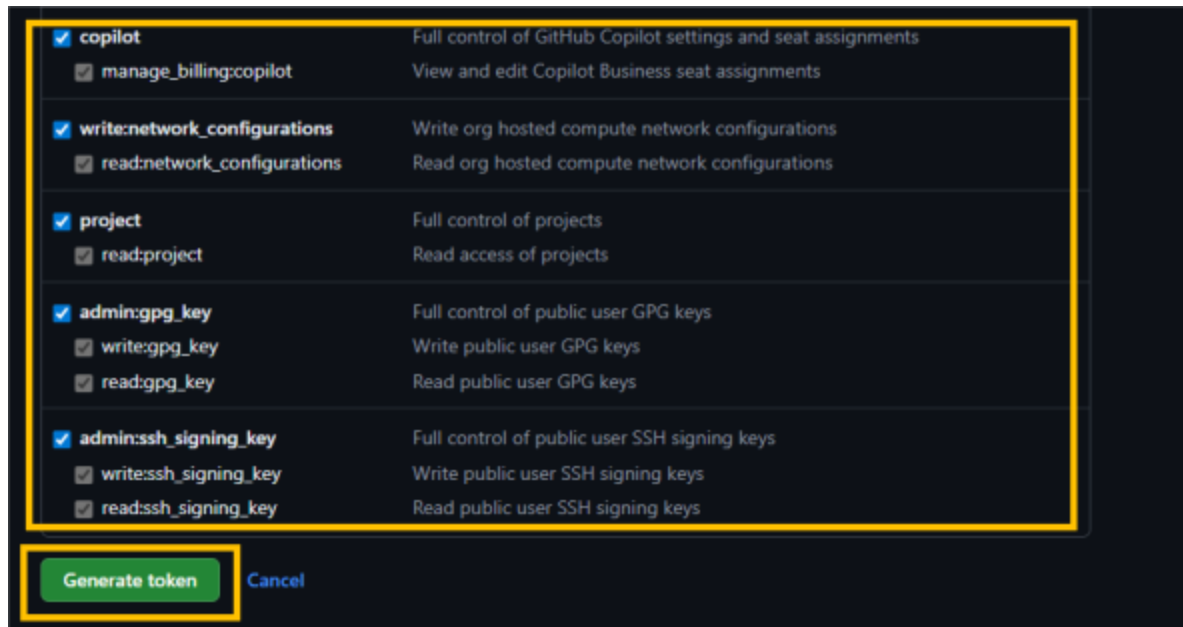
The token will expire on the selected date

Select scopes

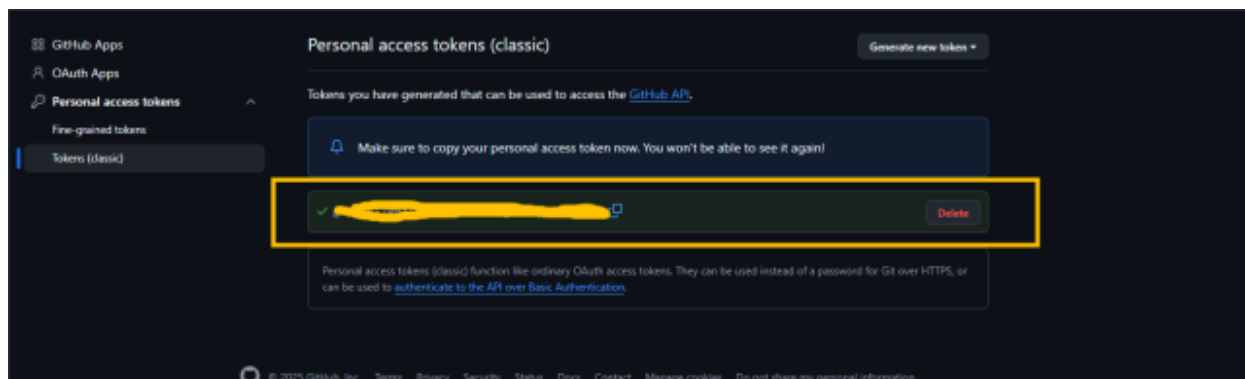
Scopes define the access for personal tokens. [Read more about OAuth scopes.](#)

☒ repo	Full control of private repositories
☒ repo:status	Access commit status
☒ repo_deployment	Access deployment status
☒ public_repo	Access public repositories
☒ repo:invite	Access repository invitations
☒ security_events	Read and write security events
☒ workflow	Update GitHub Action workflows
☒ write:packages	Upload packages to GitHub Package Registry
☒ read:packages	Download packages from GitHub Package Registry
☒ delete:packages	Delete packages from GitHub Package Registry
☒ admin:org	Full control of orgs and teams, read and write org projects
☒ write:org	Read and write org and team membership, read and write org projects
☒ read:org	Read org and team membership, read org projects
☒ manage_runners:org	Manage org runners and runner groups

After selecting the scope, click on “Generate token”.



Here is token.



You can follow this configuration.

Link: https://documentation.wazuh.com/current/cloud-security/github/m_onitoring_github-activity.html

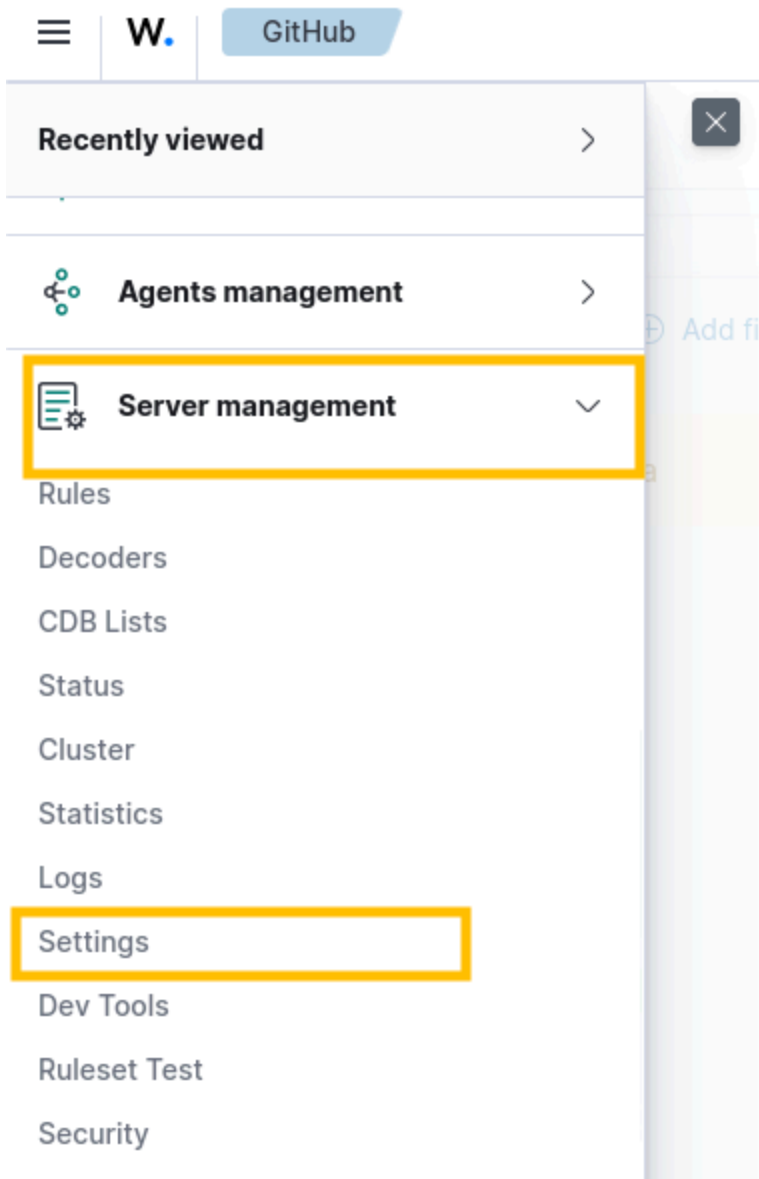
Configure Wazuh to pull GitHub logs

Perform the following steps to allow Wazuh to monitor, collect, and analyze the GitHub audit logs. You can either configure the Wazuh module for GitHub in the Wazuh server or the Wazuh agent.

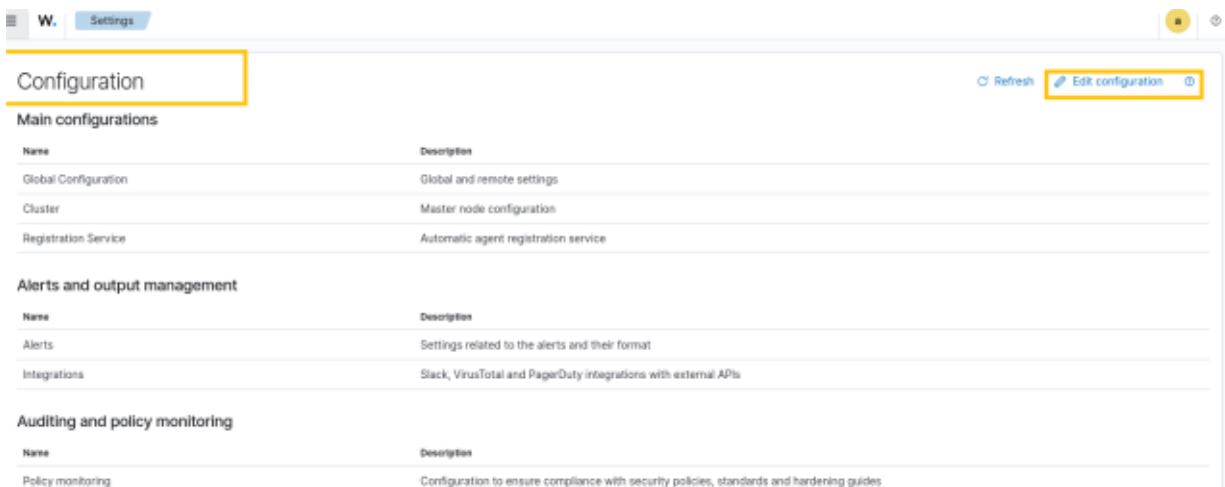
1. Append the following configuration to the `/var/ossec/etc/ossec.conf` file on the Wazuh server.

```
<ossec_config>
  <github>
    <enabled>yes</enabled>
    <interval>1m</interval>
    <time_delay>1m</time_delay>
    <curl_max_size>1M</curl_max_size>
    <only_future_events>yes</only_future_events>
    <api_auth>
      <org_name><ORG_NAME></org_name>
      <api_token><API_TOKEN></api_token>
    </api_auth>
    <api_parameters>
      <event_type>all</event_type>
    </api_parameters>
  </github>
```

In Wazuh-dashboard go to “Server Management” and select “Settings”.



Now we have to edit “ossec.conf”.



<github>

<enabled>yes</enabled>

<interval>1m</interval>

<time_delay>1m</time_delay>

<curl_max_size>1M</curl_max_size>

<only_future_events>yes</only_future_events>

<api_auth>

<org_name><ORG_NAME></org_name>

<api_token><API_TOKEN></api_token>

</api_auth>

<api_parameters>

<event_type>all</event_type>

</api_parameters>

</github>

We have to add configuration here.

< Manager configuration

Refresh XML format error Restart Manager

Edit ossec.conf of Manager Error validating XML

```
127 <!-- Slack Integration -->
128 <github>
129   <enabled>yes</enabled>
130   <interval>1m</interval>
131   <time_delay>1m</time_delay>
132   <curl_max_size>1M</curl_max_size>
133   <only_future_events>yes</only_future_events>
134   <api_auth>
135     <org_name>ORG_NAME</org_name>
136     <api_token>API_TOKEN</api_token>
137   </api_auth>
138   <api_parameters>
139     <event_type>all</event_type>
140   </api_parameters>
141 </github>
142 </integration>
```

Follow the same and add “GitHub” generated token in configuration.

< Manager configuration

Refresh XML format error Restart Manager

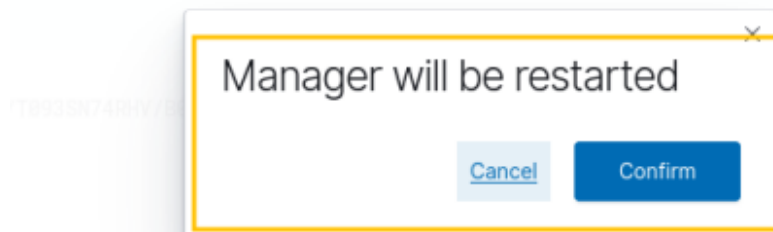
Edit ossec.conf of Manager Error validating XML

```
127 <!-- Slack Integration -->
128 <github>
129   <enabled>yes</enabled>
130   <interval>1m</interval>
131   <time_delay>1m</time_delay>
132   <curl_max_size>1M</curl_max_size>
133   <only_future_events>yes</only_future_events>
134   <api_auth>
135     <org_name>ORG_NAME</org_name>
136     <api_token>API_TOKEN</api_token>
137   </api_auth>
138   <api_parameters>
139     <event_type>all</event_type>
140   </api_parameters>
141 </github>
142 </integration>
```

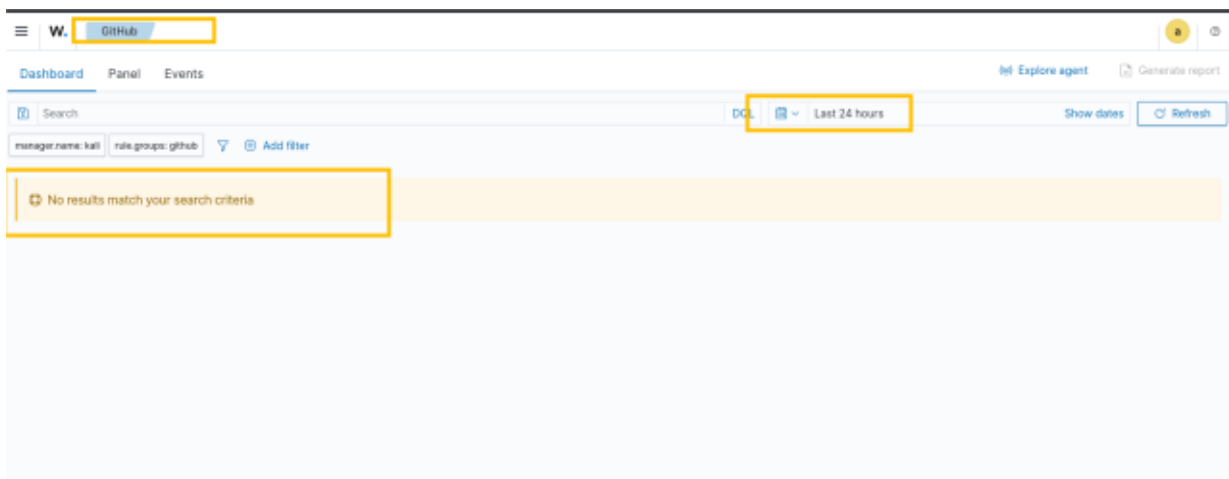


Save the configuration and restart Wazuh-manager.

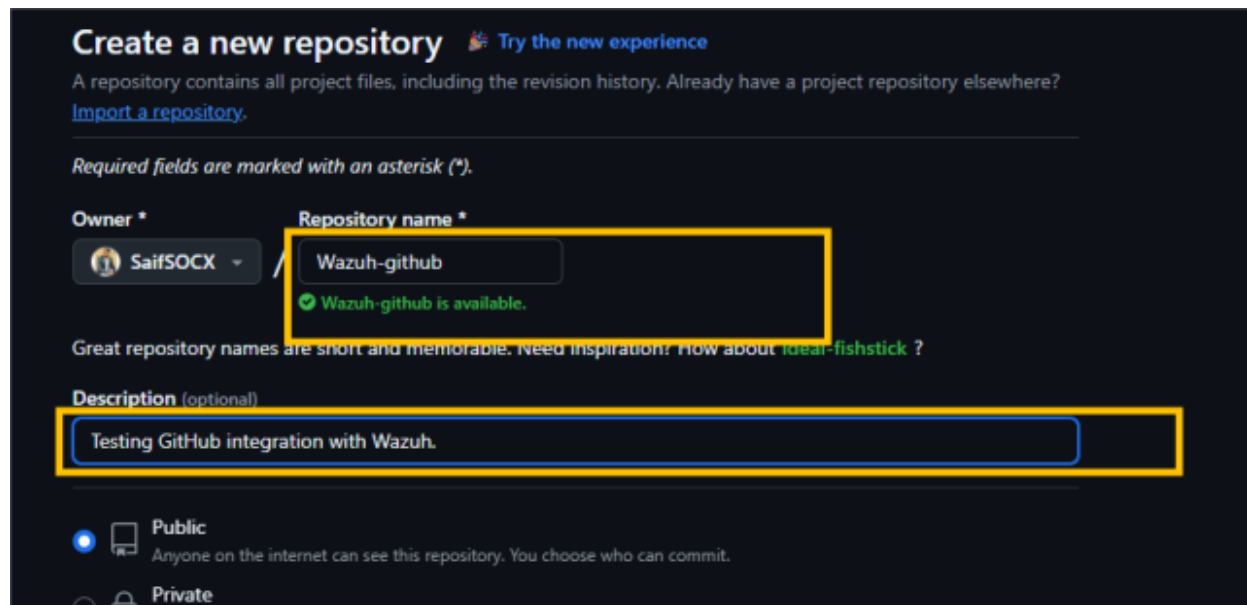
Restarting Wazuh-manager.



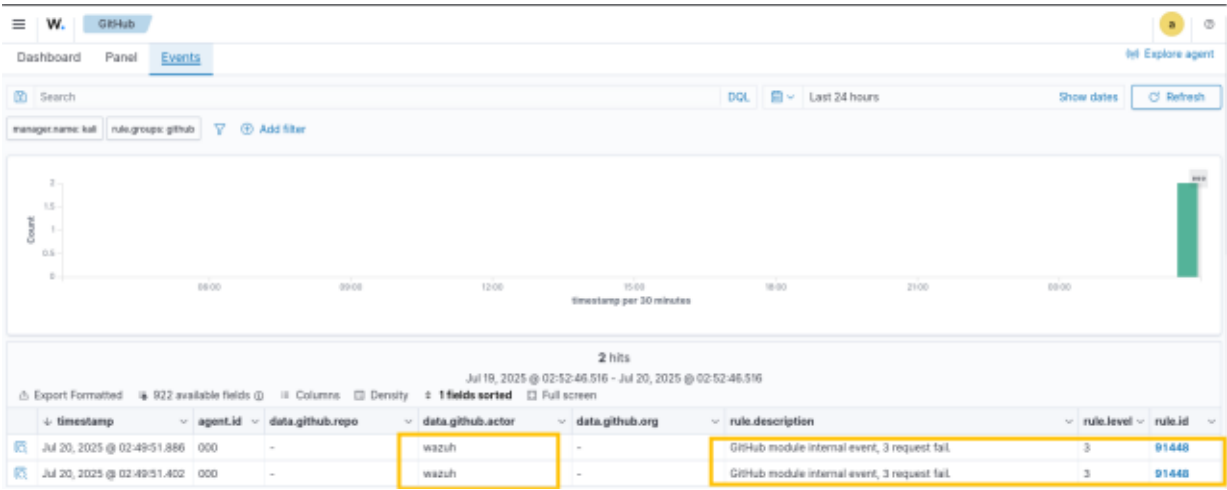
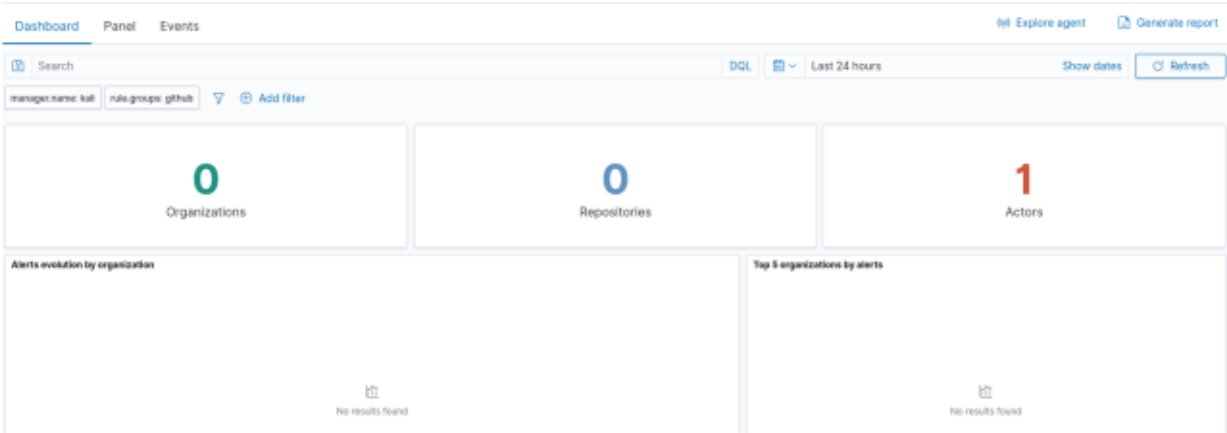
After restarting, there is no logs available.



Now we have to generate any type of events for confirm integration.



Here you can see event.



Logs

Table JSON	
@timestamp	Jul 26, 2025 @ 02:49:51.896
_index	wazuh-alerts-4.x-2025.07.26
agent.id	000
agent.name	kali
data.github.actor	wazuh
data.github.event_type	web
data.github.organization	Scif50CX
data.github.response	{"message": "Not Found", "documentation_url": "https://docs.github.com/rest/orgs/orgs#get-the-audit-log-for-an-organization", "status": "404"}
data.integration	github
decoder.name	json
id	1752994191.22862
input.type	log
location	github
manager.name	kali
rule.description	GitHub module internal event, 3 request fail.
# rule.firedtimes	2
rule.name	github_000

We can get “Actors” “Organizations” “Repositories” “Action” events. My recommendation follow the link: <https://documentation.wazuh.com/current/cloud-security/github/monitoring-github-activity.html> to generates events.

Dashboard Events	
Actor: 0	Organization: 0
Repository: 0	Action: 0
Last 24 hours	
Show dates Refresh	
manager.name: kali rule.groups: github Add filter	
Advanced filters	
Actors	Organizations
Actor	Organization
Count	Count
2	No items found
< 1 >	
Repositories	Actions
Repository	Action
Count	Count
No items found	No items found

SUMMARY:

integrating Wazuh with GitHub, organizations can significantly enhance their DevSecOps capabilities, ensuring that their code repositories remain secure and compliant with organizational standards.